

Bachelor in Computer Applications

Course Title: **Digital Logics**

Code No: Semester: **1**

Full Marks: **60** Pass Marks: **24** Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Note: Group A (Multiple Choice Questions) is not available from the source. The paper begins with Group B.

Group B

2. Define combinational logic circuit. Design a combinational circuit whose input is a four-bit number and output is 2's complement of the input number.
3. What is ring counter? Explain ring counter with diagram, timing sequence and timing diagram.
4. Convert (1011 110) into decimal and hexadecimal.
5. Subtract (1001.11 - 1010.10) using 2's and 1's complement.
6. Express the given function in sum of minterms. $F = y'z + wxy' + wzx' + w'x'z'$
7. Simplify the Boolean function using don't care conditions d, In sum of products and product of sums form. $F(A, B, C, D) = \pi(1,3,7,8,12)$ and $\pi d(5,10,13,14)$
8. Implement a full subtractor with two half subtractors and one OR Gate.
9. Define SR latch with logic diagram and truth table.
10. Compare the logic of synchronous counter and ripple counter.
11. Define shift register with its types.
12. Write short notes on: a) Parity generator b) State diagram